

Atomic and Molecular Calculations with Explicitly Correlated Gaussians: An Efficient Open Implementation in Julia

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Abstract

We present `ECG_Julia`, an open, modular implementation in the Julia programming language of the variational method of explicitly correlated Gaussians (ECGs) for non-relativistic, finite-nuclear-mass atomic and molecular bound-state calculations. The code follows the formalism reviewed by Bubin, Pavanello, Tung, Sharkey and Adamowicz [1] and the matrix-element algebra of Kinghorn [4], Bubin and Adamowicz [3], and Sharkey and Adamowicz [2]. Four basis-function families are supported: two real correlated Gaussians for $L = 0$ states (`RGL0`), two complex correlated Gaussians for $L = 0$ states (`CGL0`), the $L = 0$ nitrogen basis with a cubic angular prefactor (`N_Pvec/N_C_Pvec`), and two real correlated Gaussians for $L = 1$ states (`RGL1`). The package combines a JSON-driven run configuration, an automatically-differentiated analytic energy gradient, a symmetry-term truncation/ramp scheme for accelerating the non-linear optimization of large permutation operators, and an optional text-file bridge to the reference Fortran program `ATOM-MOL-NONBO`. The primary contribution is the implementation itself—its modular software architecture, its forward-mode automatic-differentiation design, and its on-the-fly construction and handling of the permutational-symmetry operator—released as open-source software so that it can be inspected, reused and extended. To verify correctness across all four models we reproduce published ground-state energies for HD^+ , D_3^+ , H_3^+ , ^9Be , He , ^7Li , ^{14}N , and the positronium molecule Ps_2 ; these calculations serve as validation of the code rather than as new or best-available reference data.

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Executive summary

Explicitly correlated Gaussian (ECG) basis functions provide some of the most accurate non-relativistic energies available for small atoms and molecules because the interparticle distances enter the basis explicitly, and because the Born–Oppenheimer separation can be dropped entirely (all particles, nuclei and electrons, are treated on the same footing) [1]. The price is a demanding non-linear optimization of the Gaussian exponent matrices and an expensive permutational-symmetry projection.

This work re-implements the established ECG methodology in Julia, with three aims: (i) a clean, configuration-driven, reproducible code base in which each physical model is a self-contained matrix-element module; (ii) a faster nonlinear optimization, obtained from an exact analytic energy gradient generated by forward-mode automatic differentiation and from a multi-fidelity truncation/ramp of the symmetry operator; and (iii) interoperability with the reference Fortran program `ATOM-MOL-NONBO` through a plain text-file interface, so that results can be cross-checked. The deliverable is therefore the open implementation itself—a reusable, extensible code base for ECG calculations—of which the benchmark energies reported below are a verification rather than the end product. Section 1 summarizes the ECG formalism and the four supported models; Section 2 describes the Julia implementation; Section 3 verifies the implementation against published energies; Section 4 concludes. Appendices A–C give the Hamiltonian and permutational symmetry, the solution of the secular equation (including the analytic/automatic-differentiation gradient), and the `vec`/`vech` matrix operators.

1 Overview of atomic and molecular calculations with explicitly correlated Gaussians

For the authoritative and comprehensive account of the method we refer the reader to Bubin, Pavanello, Tung, Sharkey and Adamowicz, *Born–Oppenheimer and Non-Born–Oppenheimer, Atomic and Molecular Calculations with Explicitly Correlated Gaussians*, Chem. Rev. **113**, 36–79 (2013) [1]. The present section recalls only what is needed to define the four models implemented in `ECG_Julia`.

For an n -pseudoparticle system (after separation of the center-of-mass motion) with internal Cartesian coordinates collected in the $3n$ -vector $\mathbf{r}$, the variational wave function is a linear combination of symmetry-projected ECGs,

$$\psi(\mathbf{r}) = \sum_{k=1}^K c_k \hat{Y} \phi_k(\mathbf{r}), \quad (1)$$

where $\hat{Y}$ is the permutational-symmetry projector (a linear combination of permutation operators; Appendix A), the c_k are linear variational coefficients, and the ϕ_k are correlated Gaussians whose

exponent matrices contain the nonlinear variational parameters. Minimizing the Rayleigh quotient with respect to the c_k leads to the generalized symmetric/Hermitian eigenproblem

$$(\mathbf{H} - \varepsilon \mathbf{S}) \mathbf{c} = \mathbf{0}, \quad \mathbf{H}_{kl} = \langle \phi_k | \hat{H} \hat{Y}^\dagger \hat{Y} | \phi_l \rangle, \quad \mathbf{S}_{kl} = \langle \phi_k | \hat{Y}^\dagger \hat{Y} | \phi_l \rangle, \quad (2)$$

solved repeatedly inside an outer optimization of the nonlinear parameters (Appendix B). The four models differ in the form of ϕ_k .

1.1 Two real correlated Gaussians for states with rotation quantum number $L = 0$ (module RGL0)

For $L = 0$ states (all particles in s -states) the spatial basis function has only a radial exponential part,

$$\phi_k(\mathbf{r}) = \exp[-\mathbf{r}' \overline{\mathbf{A}}_k \mathbf{r}], \quad k = 1, \dots, K, \quad (3)$$

where the prime denotes transposition. The $3n \times 3n$ real symmetric exponent matrix factorizes as $\overline{\mathbf{A}}_k = A_k \otimes I_3$, with I_3 the 3×3 identity, $\otimes$ the Kronecker product, and A_k an $n \times n$ symmetric matrix. To guarantee a positive-definite quadratic form, A_k is held in Cholesky-factored form $A_k = L_k L'_k$ with L_k an $n \times n$ lower-triangular matrix, so that

$$\phi_k(\mathbf{r}) = \exp[-\mathbf{r}' (L_k L'_k \otimes I_3) \mathbf{r}]. \quad (4)$$

The nonlinear variational parameters are the lower-triangular elements of the L_k (the vech L_k ; Appendix C). This is the ECG_Matelem_RGL0 module.

1.2 Two complex correlated Gaussians for states with rotation quantum number $L = 0$ (module CGL0)

Complex exponent matrices add variational flexibility [1]. The basis function reads

$$\phi_k(\mathbf{r}) = \exp[-\mathbf{r}' \overline{\mathbf{C}}_k \mathbf{r}] = \exp[-\mathbf{r}' (\overline{\mathbf{A}}_k + i \overline{\mathbf{B}}_k) \mathbf{r}], \quad (5)$$

where $\overline{\mathbf{A}}_k = A_k \otimes I_3$ and $\overline{\mathbf{B}}_k = B_k \otimes I_3$ with A_k, B_k real symmetric $n \times n$ matrices, $A_k = L_k L'_k$ as before. This is the ECG_Matelem_CGL0 module.

1.2.1 Nitrogen with rotation quantum number $L = 0$ (modules N_Pvec, N_matr, N_C_Pvec)

For the 4S ground state of nitrogen, with three p -electrons coupled to $L = 0, M = 0$, the basis function carries a cubic angular prefactor [2]:

$$\begin{aligned} \psi_f = & \left\{ (x_{j_f} y_{i_f} - x_{i_f} y_{j_f}) z_{k_f} + (x_{i_f} y_{k_f} - x_{k_f} y_{i_f}) z_{j_f} \right. \\ & \left. + (x_{k_f} y_{j_f} - x_{j_f} y_{k_f}) z_{i_f} \right\} \exp[-\mathbf{r}' \mathbf{A}_f \mathbf{r}]. \end{aligned} \quad (6)$$

This prefactor has six unique terms, so each Hamiltonian or overlap matrix element requires the evaluation of 36 elemental integrals.

Permutational symmetry. Symmetry adaptation is enforced by projecting with the Young operator $\hat{Y}$ (Appendix A):

$$\psi_k = \hat{Y} \Phi_k = \sum_P \chi_P r_1^{m_k} \exp[-\mathbf{r}' (\tau'_P L_k L'_k \tau_P \otimes I_3) \mathbf{r}], \quad (7)$$

the sum running over permutations of identical-particle labels, with permutation matrices τ_P . For nitrogen the complete operator $\hat{Y}^\dagger \hat{Y}$ has $n! = 7! = 5040$ terms.

Overlap, potential and kinetic matrix elements. With A_{fg} the combined exponent matrix and $\eta_x \equiv (A_{fg}^{-1})_{(x_f, P \text{ vec}(x_g))}$ (and analogously for y, z), the overlap is [2]

$$\langle \phi_f | \phi_g \rangle = \frac{1}{8} \pi^{3n/2} |A_{fg}|^{-3/2} \eta_x \eta_y \eta_z, \quad (8)$$

the electron–nucleus / electron–electron potential element is

$$\begin{aligned} \left\langle \phi_f \left| \frac{1}{r_{i'j'}} \right| \phi_g \right\rangle = & \frac{1}{4} \pi^{(3n-1)/2} |A_{fg}|^{-3/2} \lambda^{-1/2} \left\{ \eta_x \eta_y \eta_z - \frac{\lambda^{-1}}{3} (\eta_x \eta_y \eta_{zJ} + \eta_x \eta_y J \eta_z + \eta_{xJ} \eta_y \eta_z) \right. \\ & \left. + \frac{\lambda^{-2}}{5} (\eta_x \eta_y J \eta_{zJ} + \eta_{xJ} \eta_y \eta_{zJ} + \eta_{xJ} J \eta_y \eta_z) - \frac{\lambda^{-3}}{7} \eta_{xJ} J \eta_y J \eta_{zJ} \right\}, \end{aligned} \quad (9)$$

and the kinetic element is

$$\begin{aligned} \langle \nabla_{\mathbf{r}} \phi_f | \mathbf{M} | \nabla_{\mathbf{r}} \phi_g \rangle = & \frac{1}{4} \pi^{3n/2} |A_{fg}|^{-3/2} \left\{ 3\tau \eta_x \eta_y \eta_z \right. \\ & + (M_{(x_f, P \text{ vec}(x_g))} + 2\tau_x - (A_{fg}^{-1} A_f M)_{(x_f, P \text{ vec}(x_g))} - (M A_g A_{fg}^{-1})_{(x_f, P \text{ vec}(x_g))}) \eta_y \eta_z \\ & \left. + (x \rightarrow y) \eta_x \eta_z + (x \rightarrow z) \eta_x \eta_y \right\}. \end{aligned} \quad (10)$$

Three modules implement these elements: `N_Pvec` and `N_matr` (real Gaussians) and `N_C_Pvec` (complex). With the 36 elemental integrals carried by each of the $n! = 5040$ permutation terms, evaluating a single nitrogen Hamiltonian or overlap matrix element requires the elemental-integral formula to be applied 180,240 times [2], so minimizing the work per permutation is essential. Following the permutational-symmetry scheme of Sharkey and Adamowicz [2] (Sec. IV), the `Pvec` variants avoid forming permutation-matrix products altogether: the forwards-direction permutation ($P_g v = v' P'_g$) is stored in the first half and the backwards-direction permutation ($P'_g v = v P'_g$) in the second half of a $2n$ -component index vector `pvec` (Appendix C), so a permuted width-matrix element is obtained by an index gather,

$$(P_g A P'_g)_{(s,t)} = A_{(\text{pvec}(s), \text{pvec}(t))}, \quad (P'_g A P_g)_{(s,t)} = A_{(\text{pvec}(s+n), \text{pvec}(t+n))}, \quad (11)$$

rather than by multiplying matrices — “as long as we know the order of the particles, we know the order of the permuted matrix” [2]. This reduces forming $P'_g A_g P_g$ from $2n^3$ to n^2 operations (for nitrogen, $686 \rightarrow 49$) and permuting a $3n$ -vector from $(3n)^2$ to n ($441 \rightarrow 7$) — the principal per-permutation saving that `ECG_Julia` inherits.

1.3 Two real correlated Gaussians for states with rotation quantum number $L = 1$ (module `RGL1`)

For $L = 1$ states (one particle in a p -state, the rest in s -states) the basis function carries a single linear prefactor [3]:

$$\phi_k(\mathbf{r}) = z_{m_k} \exp[-\mathbf{r}' \bar{\mathbf{A}}_k \mathbf{r}] = z_{m_k} \exp[-\mathbf{r}' (L_k L'_k \otimes I_3) \mathbf{r}], \quad (12)$$

where m_k is an integer index depending on k . The energy and energy-gradient matrix elements for N -particle $L = 1$ complex Gaussians are those of Bubin and Adamowicz [3]. This is the `ECG_Matelem_RGL1` module; we use it below for the ground state of the positronium molecule Ps_2 in the $L = 1$, negative-parity state.

2 An efficient implementation in Julia

ECG_Julia is organized as a small set of modules, and rests on three design choices that together constitute the contribution of this work: a *modular software architecture* in which each physical model is a self-contained matrix-element module behind a single uniform interface; a *generic-programming / automatic-differentiation design* that yields exact analytic energy gradients without hand-coded derivatives; and an *on-the-fly construction of the permutational-symmetry operator* from a human-readable Young-operator string. The Julia language [12] provides compiled performance with a high-level, generic programming model; the latter is essential to the automatic-differentiation gradient described below. We describe each module in turn, grouping the discussion under these three themes.

ECG_Param. Defines the run parameters and the default-parameter structure `param0` (an `Init_param` instance): system size, basis size, target energy, optimization controls, and the model selector `MatElem_method` ($\in \{\text{RGL0, CGLO, RGL1, N_Pvec, N_matr, N_C_Pvec}\}$). Recent fields control the optimization accelerators: `coeff_threshold` and `coeff_ramp_tol` (symmetry-term truncation and ramp), `analytic_grad` (analytic energy gradient), and the trial-generation controls `gen_NonlinParam`, `RG_p1`, `RG_s1`, `RG_s2`.

ECG_Config. Reconstructs an entire run from a single JSON file (`ecg_config.json`) with three sections: `setup` (the module and data files copied into the working directory, and stale files removed, using native-Julia file operations), `ECG_Param` (module-level settings), and `param0` (fields of the `Init_param` instance). This replaces ad-hoc post-include mutation of globals and makes runs reproducible and machine-writable.

ECG_Init. Initializes the data structures, reduced-mass and charge data, the transposition matrices, and the symmetry (Y and $Y^\dagger Y$) operators. The operators are read from reference files when present, and otherwise computed on the fly by the module `SymmetryOperators` directly from the Young-operator string (e.g. $(1+P12)(1-P23)$) and the particle count. The module also generates the Basis-Function Prefactor Index (BFPI) table for the $L = 0$ atomic prefactors and defines `set_parvec`, which assembles the 36-row index array consumed by the nitrogen matrix-element kernels.

ECG. Assembles the Hamiltonian and overlap matrices (`compute_H_S`), solves the secular equation (`solve_sec`, with a Bunch–Kaufman/inverse-iteration path `GSEPIIS` and a generalized-eigensolver path), generates trial nonlinear parameters (`stoch_nlp`), performs cyclic nonlinear optimization (`optim_nlp`, built on `Optim.jl`), and orchestrates the action list of a run (`do_action`). It includes the relevant `ECG_MatElem` module at load time according to `MatElem_method`.

Two accelerators are implemented here. First, a *symmetry-term truncation and ramp*: during the inner BFGS search, $Y^\dagger Y$ terms whose coefficient magnitude falls below `coeff_threshold` are skipped, giving a cheaper partially-symmetrized surface, while every accepted or reported energy is recomputed with the full operator; an automatic ramp lowers the threshold tier by tier (eventually to the full operator) as the per-cycle improvement decays. For nitrogen, whose 5040-term operator has coefficient magnitudes in $\{96, 192, 960\}$, this concentrates early effort on the 240 dominant terms. Second, an *analytic energy gradient* (Appendix B): when `analytic_grad` is set, `optim_nlp` passes `loss_grad!` to the optimizer instead of relying on finite differences.

ECG_Fortran. Groups the functions that drive and read back the reference Fortran program ATOM-MOL-NONBO [10] through a text-file interface only: parameters and matrices are exchanged through plain files and the executable is launched as a subprocess. This enables independent cross-checks of the Julia energies.

ECG_Matelem family. Each model is a self-contained matrix-element module named `ECG_Matelem_<Model>` (e.g. `ECG_Matelem_CGL0`, `ECG_Matelem_N_Pvec`), included by ECG at load time according to `MatElem_method`. Each exposes a single `MatrixElements` function that returns the normalized H_{kl} , S_{kl} , and (optionally) their derivatives D_k , D_l with respect to the nonlinear parameters. For the nitrogen modules the derivatives are obtained by forward-mode automatic differentiation of a single type-generic kernel (`ForwardDiff.jl` [11]); seeding $\text{vech } L_k$ with dual numbers propagates the derivatives through the matrix build, the inverse and determinant, and the 36-term elemental sum, so a hand-coded derivative is no longer required. The analytic gradient assembled from these per-element derivatives reaches the same minimum as the finite-difference path at a fraction of the cost (Appendix B).

3 Verification of the implementation

Table 1 lists the benchmark systems and models used to exercise the implementation: the number of pseudoparticles n , the permutational-symmetry projector, the basis size K , the model (`MatElem_method`), the matrix-element module, and the companion Jupyter notebook. Each case reproduces published ground-state energies— HD^+ [10], D_3^+ and H_3^+ [8, 9], ^9Be [5], He [13], ^7Li [6], ^{14}N [2, 7], and Ps_2 [3]—which we record in the corresponding notebooks rather than tabulate here. These calculations verify that each model and the shared optimization machinery are correct; they are not presented as new or best-available reference values. The runs are short demonstrations (about four optimization cycles on a laptop, whereas Ref. [2], Sec. VI, notes that “about 1000 cycles” are needed for satisfactory convergence), so the obtained energies lie above the published values by an amount consistent with the limited optimization rather than any deficiency in the matrix elements.

System	n	Symmetry projector	K	Model	Notebook
HD^+	3	(P11)	10	CGLO	HD+_CGL0.ipynb
D_3^+	5	(1+P12)(1+P13+P23)(1+P45)	70	CGLO	D3+_CGL0.ipynb
H_3^+	4	(1+P12)(1+P13+P23)(1+P45)	10	CGLO	H3+_CGL0.ipynb
$^9\text{Be}(2^1S)$	5	(1-P24)(1-P35)(1+P23)(1+P45)	50	RGL0	Be_RGL0.ipynb
$^9\text{Be}(2^1S)$	5	(1-P24)(1-P35)(1+P23)(1+P45)	6	CGLO	Be_CGL0.ipynb
$\text{He}(1^1S)$	3	(1+P23)	30	RGL0	He_RGL0.ipynb
$\text{He}(1^1S)$	3	(1+P23)	30	CGLO	He_CGL0.ipynb
$^7\text{Li}(2S)$	4	(1+P12)(1-P23)	6	RGL0	Li_RGL0.ipynb
$^{14}\text{N}(4S)$	8	Young $\hat{Y}^\dagger \hat{Y}$ (7! terms)	16	N_Pvec	N_Pvec.ipynb
$\text{Ps}_2(L=1)$	4	(1-P13P24)(1+P12)(1+P34)	100	RGL1	Ps2_RGL1.ipynb

Table 1: Benchmark systems and models demonstrated with ECG_Julia: the number of pseudoparticles n , the permutational-symmetry projector, the basis size K , the model (`MatElem_method`), and the companion Jupyter notebook. The obtained ground-state energy for each case, compared with published values, is reported in the corresponding notebook.

Optimization-machinery check. As an independent check of the optimization machinery, a three-function `N_Pvec` run with the analytic gradient and the $960 \rightarrow 192 \rightarrow 0$ truncation ramp converges cleanly, reaching $E = -52.627\,624$ a.u. with a generalized-eigenproblem residual $\max |(\mathbf{H} - \epsilon \mathbf{S})\mathbf{c}| \approx 10^{-14}$. Together with Table 1, these runs validate, across all four matrix-element models, both the matrix-element algebra and the optimization machinery.

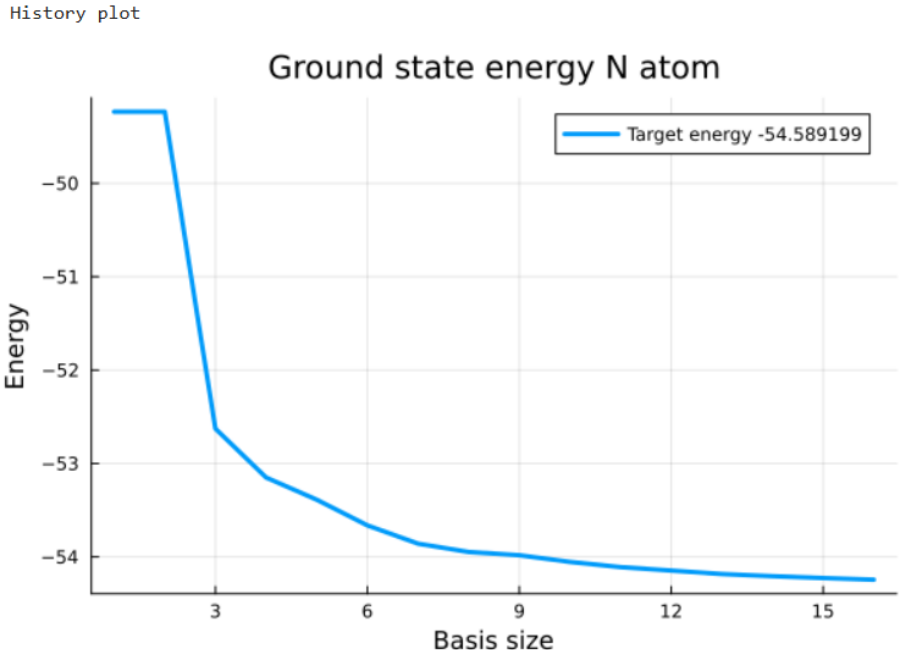


Figure 1: Convergence of the $^{14}\text{N } 4S$ ground-state energy (`N_Pvec`) with basis size K toward the target energy $-54.589\,199$ a.u. The energy decreases monotonically as basis functions are added; the value at $K = 16$ (-54.2445 a.u.) remains above the reference because of the short (~ 4 -cycle) optimization used here (see text).

4 Conclusion

We have presented an open, modular Julia implementation of the explicitly correlated Gaussian method that covers real and complex $L = 0$ Gaussians, the $L = 0$ nitrogen basis with a cubic angular prefactor, and $L = 1$ Gaussians. The implementation itself is the principal contribution: a reproducible JSON-driven configuration system; a modular matrix-element architecture in which each physical model sits behind a single uniform interface; an exact analytic energy gradient generated by automatic differentiation (eliminating hand-coded derivatives and the cost of finite differences); and an on-the-fly construction of the permutational-symmetry operator from a human-readable Young-operator string, together with a symmetry-term truncation/ramp scheme that concentrates optimization effort on the dominant permutation terms. The correctness of the code is established by reproducing published ground-state energies for HD^+ , D_3^+ , H_3^+ , ^9Be , He , ^7Li , ^{14}N , and Ps_2 , and the text-file bridge to `ATOM-MOL-NONBO` allows independent cross-checks of the energies.

Future work includes a fuller parallelization of the symmetry-term sums over the Hamiltonian, overlap and gradient matrix elements, and an evaluation of batched/accelerator implementations of the matrix-element kernels for larger basis sets, where the permutational sum dominates the cost.

Code availability. ECG_Julia is released as open-source software, together with the JSON configuration files that reproduce every entry of Table 1, so that the results reported here can be regenerated and the implementation reused and extended. *[Repository URL and open-source license to be inserted on release.]*

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A Hamiltonian and permutational symmetry

(After Sharkey and Adamowicz [2].)

For an n -electron atomic system with only s electrons the basis function is spherically symmetric and has the ECG form

$$\psi_f = \exp[-\mathbf{r}'(A_f \otimes I_3)\mathbf{r}], \quad (13)$$

with A_f an $n \times n$ symmetric matrix. The quadratic form $\mathbf{r}'(A_f \otimes I_3)\mathbf{r}$ includes all one- and two-pseudoelectron exponential couplings. For three p -electrons coupled to $L = 0$, $M = 0$ the angular part is $(x_j y_i - x_i y_j)z_k + (x_i y_k - x_k y_i)z_j + (x_k y_j - x_j y_k)z_i$, and the matrix elements are derived from the general cubic-prefactor ECG $\phi_f = (\mathbf{r}'\mathbf{v}_f^x)(\mathbf{r}'\mathbf{v}_f^y)(\mathbf{r}'\mathbf{v}_f^z) \exp[-\mathbf{r}'\mathbf{A}_f\mathbf{r}]$.

The permutational symmetry is enforced by projecting each basis function with a symmetry-projection operator $\hat{Y}$:

$$\psi_k = \hat{Y} \Phi_k = \sum_P \chi_P r_1^{m_k} \exp[-\mathbf{r}'(\tau_P' L_k L_k' \tau_P \otimes I_3)\mathbf{r}]. \quad (14)$$

For n fermions with total spin s , the symmetry quantum number is $p = \frac{n}{2} - s$ and the partition $\mu = [2^p 1^{n-2p}]$ defines the Young tableau. For nitrogen (4S , three p -electrons, $s = \frac{3}{2}$, $\mu = [2^2 1^3]$) the projector is $\hat{Y} = \hat{S}\hat{A}$ with $\hat{S} = (\hat{1} + \hat{P}_{56})(\hat{1} + \hat{P}_{78})$ and

$$\hat{A} = (\hat{1} - \hat{P}_{68})(\hat{1} - \hat{P}_{57})(\hat{1} - \hat{P}_{27} - \hat{P}_{25})(\hat{1} - \hat{P}_{32} - \hat{P}_{37} - \hat{P}_{35})(\hat{1} - \hat{P}_{43} - \hat{P}_{42} - \hat{P}_{47} - \hat{P}_{45}), \quad (15)$$

so that the complete operator $\hat{P} = \hat{Y}^\dagger \hat{Y}$ has $n! = 7! = 5040$ terms.

B Solving the secular equation and the energy gradient

(After Bubin and Adamowicz [3, 1].)

With $\psi(\mathbf{r}) = \sum_k c_k \hat{Y} \phi_k(\mathbf{r})$, minimizing the energy with respect to the c_k gives the generalized eigenproblem $(\mathbf{H} - \varepsilon \mathbf{S})\mathbf{c} = 0$ with Hermitian $\mathbf{H}, \mathbf{S}$. The nonlinear parameters are optimized with the energy gradient

$$\frac{\partial \varepsilon}{\partial \alpha} = \mathbf{c}^\dagger \left(\frac{\partial \mathbf{H}}{\partial \alpha} - \varepsilon \frac{\partial \mathbf{S}}{\partial \alpha} \right) \mathbf{c}, \quad (16)$$

where α stacks the lower-triangular elements of all L_k .

Automatic differentiation of the molecular integrals. Rather than hand-coding $\partial \text{vech } \mathbf{H} / \partial \alpha$ and $\partial \text{vech } \mathbf{S} / \partial \alpha$, the per-element derivatives are obtained by forward-mode automatic differentiation [11]. The type-generic matrix-element kernel propagates dual numbers through the matrix

build, determinant, inverse and 36-term sum, so a single Jacobian of $[\mathbf{H}_{kl}, \mathbf{S}_{kl}]$ with respect to $\text{vech } L_k$ yields

$$\text{Dk} = \left(\frac{\partial \mathbf{H}_{kl}}{\partial \text{vech } L_k}, \frac{\partial \mathbf{S}_{kl}}{\partial \text{vech } L_k} \right), \quad \text{Dl} = \left(\frac{\partial \mathbf{H}_{kl}}{\partial \text{vech } L_l}, \frac{\partial \mathbf{S}_{kl}}{\partial \text{vech } L_l} \right). \quad (17)$$

For the complex module ($\mathbf{C}_k = \mathbf{A}_k - i\mathbf{B}_k$) the Jacobian of $[\text{Re } \mathbf{H}, \text{Im } \mathbf{H}, \text{Re } \mathbf{S}, \text{Im } \mathbf{S}]$ is recombined into the complex Dk, Dl .

Analytic gradient assembly. During optimization only the function i being optimized varies, so $\partial \mathbf{H}/\partial \alpha$ and $\partial \mathbf{S}/\partial \alpha$ are non-zero only in row/column i . Working with the raw (unnormalized) generalized eigenproblem $\mathbf{H}_r \mathbf{c} = E \mathbf{S}_r \mathbf{c}$ and its $\mathbf{S}_r$ -normalized ground eigenvector $\mathbf{c}$ (the eigenvalue is invariant under the diagonal normalization), the gradient is

$$\frac{\partial E}{\partial \text{vech } L_i} = 2 \text{Re} \left[\overline{c_i} \sum_l c_l (g_l^H - E g_l^S) \right], \quad (g_l^H, g_l^S) = \sum_j \text{YHYCoeff}[j] \frac{\partial (\mathbf{H}_{il}, \mathbf{S}_{il})}{\partial \text{vech } L_i}, \quad (18)$$

which reduces, for real $\mathbf{c}, \mathbf{H}, \mathbf{S}$, to $2 c_i \sum_l c_l (g_l^H - E g_l^S)$. This is implemented as the type-generic `loss_grad!` and was verified against central finite differences to a relative accuracy of $\sim 10^{-7}$ for both the real and complex nitrogen modules.

C The vec and vech operators

(After Kinghorn [4].)

The `vec` operator stacks the columns of an $n \times n$ matrix A into an n^2 -vector. The half-vectorization `vech` A stacks the lower-triangular elements into an $n(n+1)/2$ -vector; e.g. for $n = 3$,

$$\text{vech } A = (a_{11}, a_{21}, a_{31}, a_{22}, a_{32}, a_{33})'. \quad (19)$$

The nonlinear parameters of a basis function are the entries of $\text{vech } L_k$.

Pvec. To avoid explicit multiplication by sparse permutation matrices, the forward permutation ($Pv = v'P'$) is stored in the first half and the backward permutation ($P'v = vP'$) in the second half of a $2n$ -vector (`Pvec`). A permuted matrix element is then obtained by an index gather, $(PAP')_{(s,t)} = A_{(\text{pvec}(s), \text{pvec}(t))}$ and $(P'AP)_{(s,t)} = A_{(\text{pvec}(s+n), \text{pvec}(t+n))}$, which is where the nitrogen `Pvec` kernels save operations.

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